

Deep Learning for Human Activity Recognition in Older Adults Using Wearable Accelerometer Data

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Abstract

This study evaluates a small one-dimensional convolutional neural network (1D CNN) for classifying lying, sitting, standing and walking from wearable accelerometer signals in the HAR70+ dataset. The 18 participants are separated into 14 training, two validation and two test participants. Continuous windows of 100 samples are standardised using training data only. A sensor ablation compares two-sensor, lower-back-only and thigh-only inputs over three predetermined random seeds. The two-sensor model achieved mean test accuracy of 99.65% and Macro F1 of 0.9965, exceeding the mean Macro F1 of either single-sensor configuration. An additional three-fold evaluation within the original training participants yielded mean Macro F1 of 0.9517. The difference between these evaluations indicates that strong performance on two test participants provides limited evidence of generalisation. The findings support complementary sensor inputs for this task, subject to further evaluation across participants and recording conditions.

1. Introduction

Activity recognition can support the description of daily physical behaviour in older adults (Ustad et al., 2023). The choice of topic is informed by my background in geriatric nursing and interest in understanding everyday movement. The present study addresses four-class activity classification from sensor signals. Its scope is limited to activity labels; falls, disease and independent living ability are not measured outcomes.

Evaluation on participants excluded from training is essential to assess whether the model transfers beyond the people used for fitting. Splitting overlapping windows from the same person across training and test sets could produce optimistic scores. This study therefore separates participants, examines class-specific and participant-specific errors, and compares two sensor locations to assess the contribution of the second sensor.

2. Related Work and Research Questions

Ustad et al. (2023) evaluated activity recognition in fit-to-frail older adults, including participants who used walking aids. Their results show that including older-adult data in training can improve recognition compared with a model developed from younger adults. HAR70+ therefore provides a relevant dataset for this study. Published scores are not directly compared with the present results because preprocessing and evaluation protocols differ.

Ordóñez and Roggen (2016) investigated convolutional and recurrent networks for wearable activity recognition. Their DeepConvLSTM model demonstrates the use of learned features from sensor sequences. The present study uses a smaller 1D CNN to limit computational cost and permit a controlled sensor comparison. Its contribution is an evaluated application with sensor ablation, repeated seeds and participant-grouped checks.

Three research questions guide the evaluation. RQ1 asks how well a small 1D CNN classifies four activities in held-out HAR70+ participants. RQ2 asks how performance changes when both sensors are used rather than either sensor alone. RQ3 examines class confusion and variation across random seeds and participant groups.

3. Data and Preprocessing

HAR70+ contains recordings from 18 participants aged 70-95 during a semi-structured free-living protocol of approximately 40 minutes per participant. Two three-axis accelerometers were attached to the lower back and right thigh, sampled at 50 Hz, and annotated using video recordings (Logacjov & Ustad, 2023). The original 2,259,597 rows contain seven activity codes. Retaining walking (1), standing (6), sitting (7) and lying (8) gives 2,184,001 raw rows. Shuffling and stair activities are excluded rather than merged with other labels. Participant 518 is retained despite having no lying records.

Split	People	Windows	Lying	Sitting	Standing	Walking
train	14	32,203	3,189	7,624	5,114	16,276
validation	2	3,991	408	976	625	1,982
test	2	4,172	436	958	696	2,082

Table 1. Prepared continuous windows. Many windows come from each person; they are not independent participants.

Validation participants are 510 and 514; test participants are 511 and 512. The remaining 14 participants form the training set. The supplied data show no missing sensor values. The median within-person interval is 0.020 seconds, consistent with 50 Hz.

3.1. Continuous Windows and Leakage Controls

Original records are sorted by participant and timestamp before activities are excluded. A new segment begins at each activity change or interval greater than 30 milliseconds. The threshold accommodates observed timestamp rounding; all retained within-window intervals are at most 21 milliseconds. Each window contains 100 consecutive samples with stride 50, producing 50% overlap within a stable activity bout. At 50 Hz, the nominal window duration is two seconds; the first-to-last timestamp spans approximately 1.98 seconds because 100 samples contain 99 intervals.

Segmenting before filtering prevents excluded activities and recording gaps from joining separate bouts, correcting an issue in the exploratory notebook. Each window belongs to one participant, activity and continuous segment. Channel means and standard deviations are calculated from training windows only and applied unchanged to validation and test windows. Only the six acceleration channels enter the model. Participant IDs, timestamps and labels are excluded from the inputs. Pure-label window selection defines an offline stable-activity task and does not replicate an unlabelled live stream.

4. Model and Training

Layer	Configuration	Full-input output	Parameters
Input	100 time points $\times$ 6 channels	100×6	0
Conv1D + ReLU	32 filters, kernel 5, valid	96×32	992
MaxPooling1D	Pool size 2	48×32	0
Conv1D + ReLU	64 filters, kernel 3, valid	46×64	6,208
GlobalAveragePooling1D	Average across time	64	0
Dense + ReLU	32 units	32	2,080
Dense + softmax	4 activity probabilities	4	132

Table 2. Full-input model: 9,412 trainable parameters. A single-sensor model has 8,932 parameters because its first convolution receives three channels.

The convolutions learn short signal patterns, pooling reduces the time dimension, and global average pooling summarises learned features before the four-class softmax. All input variants use the same layer configuration and prepared windows. The slightly different first-layer parameter count is part of the channel-removal ablation, so it is not a perfectly parameter-matched comparison.

Training uses Adam with learning rate 0.001 (Kingma & Ba, 2015), sparse categorical cross-entropy and batch size 64. Training is limited to 30 epochs. EarlyStopping monitors validation loss with patience 5 and restores the weights from the lowest validation-loss epoch (Keras, n.d.). No class weighting or augmentation is used. Seeds 42, 123 and 2026 are fixed in advance and all runs are reported. Seed 42 is the predetermined example for figures and error analysis; it was not selected by comparing test scores.

5. Evaluation and Sensor Comparison

Macro F1 is the primary metric because it gives equal weight to each of the four class F1 scores. Accuracy, class precision and recall, and confusion matrices provide complementary measures. All nine sensor and seed combinations are trained before test evaluation with the corrected preprocessing. Architecture, training settings and stopping rules are not adjusted using these test results. Exploratory scores on the same test participants had previously been viewed, so this split is held out from fitting but cannot be described as a fully blind evaluation.

Input	Parameters	Accuracy (%)	Macro F1
Both sensors	9,412	99.65 ± 0.23	0.9965 ± 0.0023
Lower back only	8,932	86.57 ± 0.37	0.8328 ± 0.0043
Thigh only	8,932	92.31 ± 5.55	0.8621 ± 0.1070

Table 3. Mean $\pm$ sample standard deviation over seeds 42, 123 and 2026. These SDs describe training variation on the same test participants, not population confidence intervals.

Both sensors had the highest mean Macro F1 among these presets. The full-input mean Macro F1 differs from lower-back-only by +0.1637 and from thigh-only by +0.1344. The stronger single-sensor result is thigh only. These are descriptive differences from the same test windows, not evidence of statistical significance across older adults.

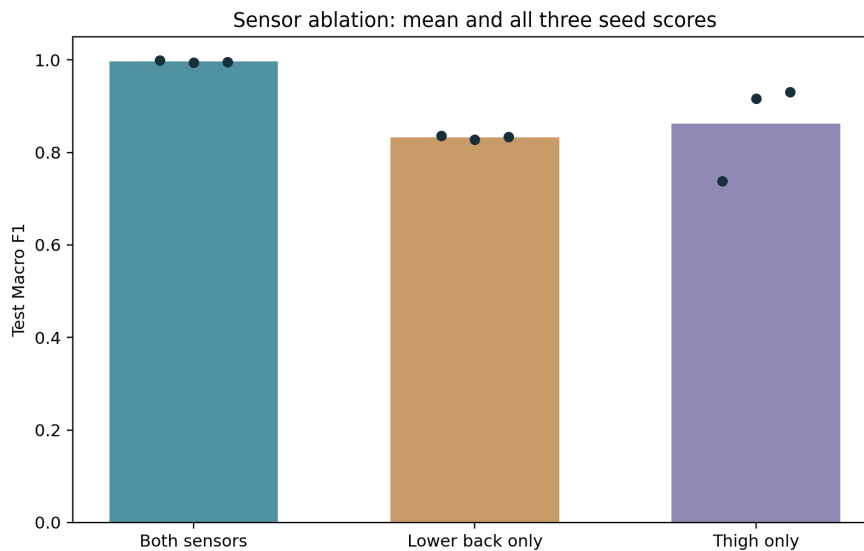


Figure 1. Mean test Macro F1 for each sensor configuration. Dots show all three predetermined seeds.

A classifier that always predicts the training-majority class, walking, achieves test accuracy of 49.90% and Macro F1 of 0.1665. This reference demonstrates the effect of class imbalance on accuracy. It provides a minimal baseline; a stronger comparison with a classical machine-learning classifier remains future work.

5.1. Training Behaviour and Grouped Robustness Check

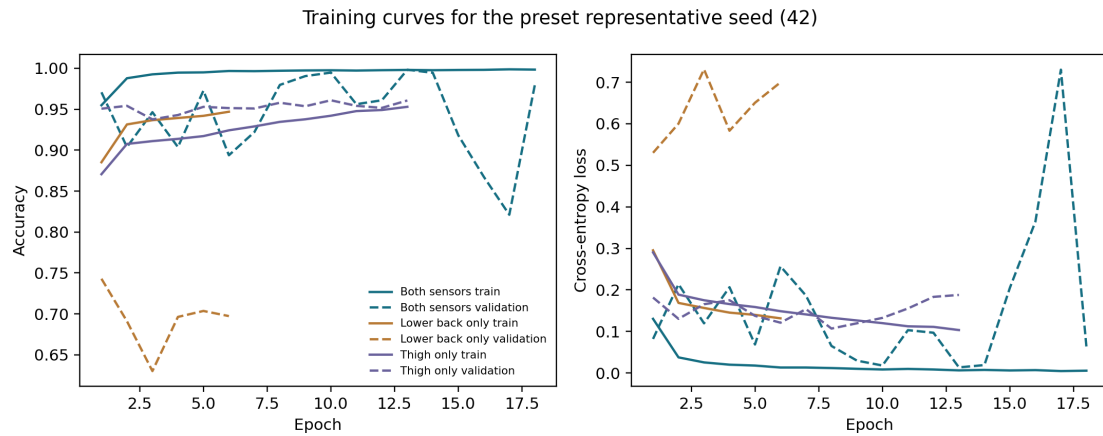


Figure 2. Preset seed-42 training and validation curves. Solid lines are training; dashed lines are validation.

The full-input runs trained for 6–18 epochs. Their restored best epochs were 13, 8, 1. Stopping duration and the restored epoch differ because patience allows several non-improving epochs. Validation-loss selection does not necessarily maximise validation Macro F1.

For an additional check, a three-fold GroupKFold partitions the original 14 training participants (Scikit-learn developers, n.d.). Each held-out fold contains four or five participants. The last two sorted participants in the remaining groups are reserved for inner early stopping; fitting uses seven or eight people. Normalisation is refitted using only those fitting participants. The full-input model uses seed 42, with unchanged architecture and settings. Original validation/test participants are excluded.

Fold	Fit / inner val / held out	Accuracy (%)	Macro F1
1	7 / 2 / 5	94.77	0.9321
2	7 / 2 / 5	98.29	0.9814
3	8 / 2 / 4	96.11	0.9417

Table 4. Different training-group sizes make these robustness scores distinct from the main 14/2/2 experiment.

Grouped Macro F1 ranges from 0.9321 to 0.9814 (unweighted fold mean 0.9517, sample SD 0.0262). This variation is a more cautious indication of generalisation than the final two-person score alone. It may reflect both differences between participants and the smaller fitting sets; it does not identify a causal reason for any particular fold's errors.

6. Error Analysis

The predetermined seed-42 two-sensor model achieves test accuracy of 99.90% and Macro F1 of 0.9990. It misclassifies 4 of 4,172 windows, with walking predicted as standing: 2; standing predicted as walking: 2. These values describe one run; Table 3 reports performance across all three seeds.

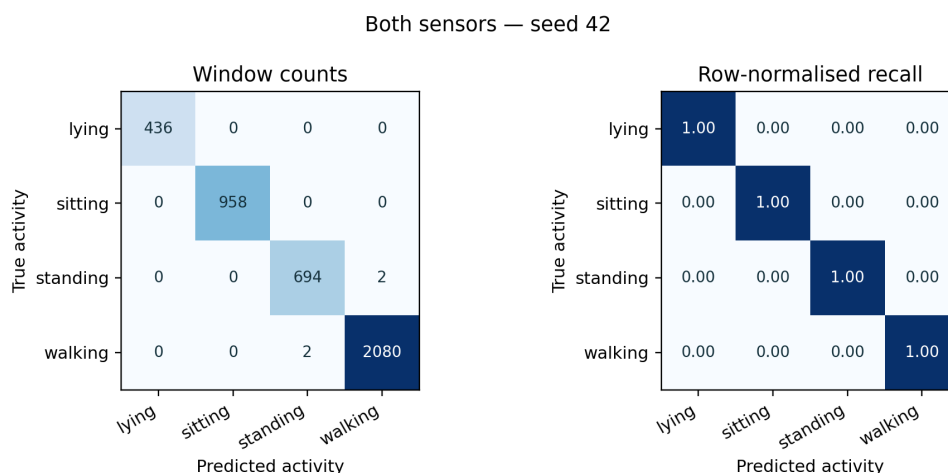


Figure 3. Counts (left) and row-normalised recall (right). Rows are true labels; columns are predictions.

Activity	Precision	Recall	F1	Windows
lying	1.0000	1.0000	1.0000	436
sitting	1.0000	1.0000	1.0000	958
standing	0.9971	0.9971	0.9971	696
walking	0.9990	0.9990	0.9990	2082

Table 5. Full-input class metrics for the predetermined seed-42 example.

The lowest class recall is standing (0.9971). Sensor signals can share similar short patterns, so confusion is plausible. However, an error label alone cannot establish whether slow movement, sensor placement or annotation caused a particular error. Video review and additional participant information would be needed to test those explanations.

6.1. Participant-Level and Specific Errors

Test participant	Windows	Accuracy (%)	Macro F1
511	2236	99.91	0.9986
512	1936	99.90	0.9992

Table 6. Seed-42 full-input model, evaluated separately on the two test participants.

Overall metrics are computed from pooled window predictions. For accuracy, participants with more windows therefore contribute more weight. Separate participant scores make this aggregation explicit. Strong performance for both participants does not remove the uncertainty associated with a two-person test sample.

The single-sensor errors help explain the ablation. For seed 42, lower-back-only sitting recall is 0.4948, while thigh-only lying recall is 0.2225. Both classes have recall 1.0000 with the full input on this split. This is consistent with complementary posture information from the two sensor locations. It is a plausible interpretation, not a proven mechanism. Thigh-only Macro F1 also varies substantially across seeds (0.7388-0.9314), so its single seed-42 result should not be treated as typical of every run.

One recorded error is participant 511, starting at 2021-05-07 10:33:51.683: true standing, predicted walking, with softmax confidence 0.843. This example shows that a confident output can still be wrong. Softmax scores were not calibrated and should not be interpreted as clinical certainty.

The accompanying error table records up to five high-confidence mistakes per sensor configuration, including participant, timestamps, ground-truth labels and predictions. Overlapping mistakes may belong to the same episode. They are therefore counted as error windows rather than independent clinical events.

7. Limitations and Future Work

First, the population sample is small: 18 participants overall and two in the final test. Repeated seeds measure optimisation variation, not uncertainty across people. Grouped validation broadens the people evaluated but still uses the same dataset. Second, excluding shuffling, stairs and mixed-label windows makes this a narrower task than recognising all daily behaviour. Segmenting by the ground-truth label is appropriate for this stable-window experiment, but would not be available during live deployment.

Third, the two-second duration, architecture, stopping rule and lack of class weighting are fixed design choices, not proven optimal choices. No systematic hyperparameter search or strong classical baseline is included. Fourth, there is no external dataset evaluation, long-term monitoring, calibration assessment or clinical outcome measurement. The study cannot show improvements in independence or fall prevention.

Future work should evaluate additional people and an independent dataset, include transition and rare-activity windows, and compare a simple engineered-feature classifier with the CNN. Window duration and stopping metrics could be studied using participant-grouped validation. Sensor burden, missing-sensor performance and probability calibration would matter before any practical application.

8. Conclusion

A small 1D CNN classifies four stable activities in HAR70+ with mean two-sensor test accuracy of 99.65% and Macro F1 of 0.9965. The two-sensor model achieved the highest mean Macro F1 across the three evaluated configurations. Class-specific errors and participant-grouped folds provide additional evidence beyond overall accuracy. The lower grouped-fold mean Macro F1 of 0.9517 indicates that performance depends on the evaluation setting. These findings apply to the specified dataset and protocol; broader generalisation and clinical usefulness require further evidence.

9. AI Tool Disclosure

ChatGPT/Codex assisted with project planning, code generation and correction, execution of experiments, figure preparation, and report drafting and language editing. No OpenCode or local LLM was used. The reported metrics were calculated by the accompanying Python/TensorFlow workflow. Dataset and reference sources were checked against their original records. I remain responsible for reviewing the submitted code, interpreting the results and explaining the work in the presentation.

10. Code and Reproducibility

The accompanying project package includes an executed notebook, equivalent training script, README, pinned direct dependencies, original data archive, trained models, learning histories, predictions and result tables. The recorded experiments used Python 3.12.14, TensorFlow 2.20.0 and Keras 3.15.1 on a Linux CPU. Deterministic operations are requested and CPU threads are limited. Numerical results may still vary with hardware or package versions.

The README documents environment setup and reproduction. Running python train.py from the project folder executes the nine sensor and seed fits and three grouped folds, writing new outputs to results_reproduced. Recorded outputs remain in results. The protocol.json file records participant lists, preprocessing settings, seeds, package versions and the input archive SHA-256. Running python verify_results.py independently recalculates the saved scores from prediction files without retraining. All tables and figures in this report refer to the recorded results.

References

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